

A+ Developer Implementation Prompt

Chicago Bare-Metal CME Futures Microstructure Research, Model-Agnostic Decision Engine, Live Execution Operations, and Production Evidence System

Purpose: provide a self-contained, quote-ready developer prompt with implementation contracts, schemas, state transitions, validation standards, production operations, and acceptance tests. Strategy selection remains evidence-driven; no arbitrary cap is placed on surviving opportunities.

1. Executive Scope

Design, quote, and implement a professional CME futures microstructure platform using Chicago/Aurora-proximity bare metal, Rithmic live market data and execution, EdgeClear broker support, Plus500US Financial Services clearing/risk, Databento GLBX.MDP3 historical MBO event windows, and HftBacktest replay. The system should operate as a research-to-production pipeline that proves which opportunities survive latency, queue position, fill realism, fees, slippage, adverse selection, and production risk constraints.

LIVE STACK

Chicago/Aurora bare metal
-> Rithmic market data + order execution
-> EdgeClear broker layer
-> Plus500US Financial Services FCM / clearing / risk
-> CME Globex

RESEARCH STACK

Databento GLBX.MDP3 MBO
-> event-window downloader
-> HftBacktest conversion
-> queue / latency replay
-> model training and validation
-> Rithmic sim shadow
-> controlled live deployment evidence

2. Non-Negotiable Design Standard

The final system is model-agnostic. Transparent models, statistical models, machine-learning models, reinforcement-learning policies, ensembles, and black-box models may be used if they are reproducible, latency-aware, risk-controlled, and validated out of sample. The implementation must not hardwire a preferred strategy style. The opportunity universe is represented as testable hypothesis families; whatever survives validation can be promoted.

The mathematical backbone should follow professional quantitative finance standards: probability, stochastic processes, statistics, econometrics/time series, numerical methods, optimization/control, finance, and robust computer systems.

3. Mathematical Decision Object

At decision time t , define F_t as all information available up to t only. Define S_t as the current market state from MBO/depth/order-flow features, L_t as latency state, Q_t as queue/fill state, C_t as cost state, I_t as inventory, and A_t as the action set. The live model must value actions, not just produce directional signals.

$A_t = \{\text{NO_TRADE, ENTER_LONG, ENTER_SHORT, ADD, REDUCE, FLATTEN, CANCEL, REPLACE}\}$

For each action a in A_t estimate:

$P(\text{direction} \mid S_t)$
 $E[\text{return} \mid S_t, a]$
 $P(\text{fill} \mid \text{queue_state}, \text{latency}, \text{order_type})$
 $E[\text{slippage} \mid S_t, a]$
 $E[\text{adverse_selection} \mid \text{fill}, a]$
 $E[\text{cost} \mid a]$
 $P(\text{stopout} \mid S_t, a)$
Expected Shortfall / tail loss
Expected net value of action a

Decision objective:

```

a* = argmax_a [
    E[PnL(a) | F_t]
    - E[cost(a) | F_t]
    - E[slippage(a) | F_t]
    - E[adverse_selection(a) | F_t]
    - latency_penalty(a)
    - risk_penalty(a)
]

```

Every live decision must output:
 selected action, expected gross edge, expected cost, expected net edge,
 fill probability, adverse-selection estimate, latency sensitivity,
 confidence score, risk score, size, max hold time, exit condition,
 kill condition, and risk status.

4. Target Labels

Build multiple target labels. Do not train only on price direction; a directionally correct model can still lose money after queue position, latency, costs, and adverse selection.

Target	Definition	Required Horizons / Axes
Forward mid return	Mid-price movement from observable state	100ms, 250ms, 500ms, 1s, 5s, 15s, 60s
Fill-adjusted PnL	Realized PnL adjusted for queue, fill, latency, spread, fees, slippage	By action, order type, event family, product
Adverse selection	Mid movement after fill relative to execution price	100ms, 500ms, 1s, 5s
Fill probability	Probability an order fills within horizon	By product, queue state, order type, latency band
Stop-out probability	Probability risk exit triggers	By regime, volatility, liquidity
Tail loss	Expected shortfall and p95/p99 loss	By event family and product
Action value	Expected value of each action	No trade, passive entry, marketable entry, cancel, replace, flatten

5. Bare-Metal Infrastructure Contract

Area	Requirement
Server type	True Chicago/Aurora-proximity bare metal; no shared VPS assumption.
Compute	Dedicated CPU, dedicated RAM, dedicated NVMe. High single-core performance preferred.
OS	Ubuntu 22.04 LTS, Ubuntu 24.04 LTS, or Debian stable.
Memory/storage	64GB RAM minimum, 128GB preferred. 1TB NVMe minimum.
Network	Stable route to Rithmic Chicago/Aurora; latency, jitter, packet loss, and p99 behavior must be measured.
Validation	Full CME session stability test, clock sync report, CPU steal = 0, packet loss = 0 during test window.

6. Low-Latency Linux Configuration

All low-latency settings must be configuration-managed and reproducible. Final CPU layout must be adjusted to actual topology.

Example core layout:

```

Core 0: OS housekeeping, SSH, monitoring
Core 1: Rithmic session / gateway adapter
Core 2: execution gateway, order state, cancel/replace manager
Core 3: risk engine / kill switch
Cores 4-7: model/strategy workers and feature computation
Remaining cores: async logging, reporting, replay/backtest jobs

```

Kernel parameter template:

```

isolcpus=2-7 nohz_full=2-7 rcu_nocbs=2-7 processor.max_cstate=0
intel_idle.max_cstate=0 amd_idle.max_cstate=0 cpuidle.off=1

```

Required controls:

performance governor, IRQ pinning away from trading cores, no GUI,
background updates disabled during market hours, chrony or PTP-grade sync,
explicit process affinity through systemd/taskset, async logging outside hot path.

7. Live Execution Architecture

Component	Implementation Contract
Rithmic Adapter	Rust, C++, or C#.NET 8. Connect to Rithmic, consume market data/depth/MBO if enabled, maintain local book state, handle order entry, cancel/replace, fills, rejects, heartbeat, reconnects, and normalized events.
Execution Gateway	Receive normalized market events and approved model actions, run pre-trade checks, construct order intents, submit through Rithmic, manage working orders, reconcile fills, emergency flatten, write latency telemetry.
Model Runtime	Model-agnostic. Rust/C++/C# for hot path; Python/Numba acceptable for research or non-hottest paths. Hot path uses lock-free queues, shared memory ring buffer, Aeron, or in-process modules.
Telemetry	Redis/Valkey allowed for dashboards, async logs, and operator state. It must not sit between a live decision and order submission.
Risk Layer	Configuration-defined controls for exposure, order flow, cancel/replace, stale data, disconnects, heartbeat, margin, reconciliation, and flatten-all actions.

8. Required Event Telemetry

Every decision and order event must carry a deterministic event ID, model ID, session ID, product, and timestamp chain.

Required timestamps:

market_data_exchange_timestamp
local_market_data_receive_timestamp
book_update_timestamp
feature_update_timestamp
model_decision_timestamp
risk_check_start_timestamp
risk_check_pass_timestamp
order_constructed_timestamp
order_send_timestamp
rithmic_ack_timestamp
exchange_ack_timestamp if exposed
fill_timestamp
cancel_send_timestamp
cancel_ack_timestamp
replace_send_timestamp
replace_ack_timestamp
reject_timestamp
disconnect_timestamp
reconnect_timestamp

Reports by:

product, model, event type, time of day, order type, side,
volatility regime, liquidity regime, p50, p90, p95, p99, worst case.

9. Historical Data Scope

Study layer	Date range	Purpose
Primary study	2018 through present	Main model discovery, confirmation, holdout, and regime coverage.
Expansion study	June 2017 through present where data quality is acceptable	Maximize clean modern MBO coverage.
Discovery	2018-2020	Feature/model discovery.
Confirmation	2021-2022	Confirm results outside discovery.
Holdout	2023-2024	Final historical holdout before recent regime.
Recent holdout	2025 through present	Current structure validation.

Study layer	Date range	Purpose
Live/sim shadow	Latest 60 CME trading days	Production reality check against Rithmic sim/live telemetry.

10. Databento Data Plan

Dataset: GLBX.MDP3

Schema: mbo

Primary symbols: MES.v.0, MNQ.v.0, ES.v.0, NQ.v.0

Macro confirmation: ZN.v.0, ZB.v.0

Optional expansion: RTY.v.0, M2K.v.0, CL.v.0, GC.v.0

For every continuous futures request, store:

requested continuous symbol, resolved raw contract, instrument ID,
roll date, start/end timestamp, event_id, file path, request hash.

Contract-roll controls:

verify no duplicate roll windows, no overlapping resolved contracts,
store roll metadata, and separate roll-period behavior from event alpha behavior.

11. Required Event Windows

Event family	Anchor	Tight	Main	Context	Symbols
CPI	08:30 ET	08:29:30-08:35 ET	08:25-08:45 ET	08:00-09:30 ET	MES, MNQ, ES, NQ, ZN, ZB
NFP / Employment	08:30 ET	08:29:30-08:35 ET	08:25-08:45 ET	08:00-09:30 ET	MES, MNQ, ES, NQ, ZN, ZB
FOMC statement	14:00 ET	13:59:30-14:05 ET	13:55-14:15 ET	13:30-15:00 ET	MES, MNQ, ES, NQ, ZN, ZB
FOMC press conf.	14:30 ET	14:29:30-14:35 ET	14:25-14:45 ET	14:25-15:30 ET	MES, MNQ, ES, NQ, ZN, ZB
Cash equity open	09:30 ET	09:29:30-09:35 ET	09:25-09:45 ET	09:15-10:00 ET	MES, MNQ, ES, NQ
Topstep-style flat	15:10 CT	15:05-15:12 CT	14:45-15:20 CT	14:30-15:30 CT	MES, MNQ, ES, NQ
MyFunded-style flat	16:10 ET / 15:10 CT	15:05-15:12 CT	14:45-15:20 CT	14:30-15:30 CT	MES, MNQ, ES, NQ
Apex-style flat	16:59 ET / 15:59 CT	15:50-16:00 CT	15:30-16:00 CT	15:00-16:00 CT	MES, MNQ, ES, NQ
TPT-style flat	16:55 ET / 15:55 CT	15:45-16:00 CT	15:30-16:00 CT	15:00-16:00 CT	MES, MNQ, ES, NQ
Friday de-risking	Friday close	N/A	14:30-16:00 CT	14:30-16:00 CT	MES, MNQ, ES, NQ

12. Event File Contract

events.csv columns:

event_id, event_type, release_date, release_time, timezone,
window_name, start_offset_seconds, end_offset_seconds,
symbols, priority, source, notes

Example rows:

CPI_2024_09_11_TIGHT, CPI, 2024-09-11, 08:30:00, America/New_York, TIGHT, -30, 300, "MES.v.0, MNQ.v.0, ES.v.0, NQ.v.0, ZN.v.0, ZB.v.0,
tight event window"

FOMC_2024_09_18_MAIN, FOMC, 2024-09-18, 14:00:00, America/New_York, MAIN, -300, 900, "MES.v.0, MNQ.v.0, ES.v.0, NQ.v.0, ZN.v.0, ZB.v.0,
statement main event window"

TOPSTEP_2024_09_18_MAIN, PROP_FLATTEN_TOPSTEP, 2024-09-18, 15:10:00, America/Chicago, MAIN, -1500, 600, "MES.v.0, MNQ.v.0, ES.v.0,
flat window"

13. Databento Credit Mathematics

Before every request:

```
cost_i = metadata.get_cost(  
    dataset="GLBX.MDP3",  
    schema="mbo",  
    symbols=symbols_i,  
    start=start_i,  
    end=end_i  
)
```

Cost accounting:

```
total_used = sum(cost_i)  
credit_remaining = 125.00 - total_used  
duration_seconds_i = end_i - start_i  
symbol_count_i = len(symbols_i)  
cost_per_symbol_second_i = cost_i / (symbol_count_i * duration_seconds_i)  
cost_per_symbol_minute_i = cost_i / (symbol_count_i * duration_seconds_i / 60)
```

Forecasting:

```
forecast_cost = median(cost_per_symbol_minute by event_type)  
    * planned_symbol_count  
    * planned_total_minutes
```

Policy:

```
operating_cap = 112.50  
reserve = 12.50  
single_request_soft_limit = 5.00  
single_request_hard_limit = 10.00 unless explicit approval
```

14. Data Pull Sequence

Phase	Scope	Purpose
0 - Cost calibration	2023-2025: 3 CPI, 3 NFP, 3 FOMC, 3 FOMC press, 10 Topstep/MyFunded windows, 10 Apex/TPT windows, 10 cash opens; MES/MNQ/ES/NQ.	Estimate cost per symbol-minute by event type, symbol, and volatility regime.
1 - Core research	2018-present: all CPI, NFP, FOMC, FOMC press, prop flat windows, Friday late-day windows, monthly sampled cash opens; MES/MNQ/ES/NQ.	Main historical event-window study.
2 - Macro confirmation	2018-present CPI/NFP/FOMC/FOMC press with ZN/ZB added.	Rates-equity lead/lag and macro impulse confirmation.
3 - Clean MBO expansion	June 2017-present where data quality is acceptable.	Maximize regime coverage.
4 - Final holdout	2025-present and latest 60 CME trading days.	Final model approval and live/sim comparison.

15. HftBacktest Pipeline

Input: .dbn.zst / .mbo.dbn.zst

Output: .npz arrays

Conversion metadata required:

```
event_id, event_type, symbol, resolved contract, instrument ID,  
input_file, output_file, start_utc, end_utc, row_count,  
conversion_status, data_quality_status
```

Backtest models:

```
Queue: LogProbQueueModel2, SquareProbQueueModel  
Exchange: NoPartialFillExchange  
Latency: 0.5ms, 1ms, 2ms, 5ms, 10ms,  
    empirical latency model after Rithmic samples exist
```

Cost model:

```
commissions, exchange fees, Rithmic/platform fees if allocatable,  
spread crossing cost, missed-fill cost, adverse selection.
```

16. Required Feature Families

Compute each feature using only information available up to timestamp t .

- aggressor_volume_imbalance
- buy_aggressor_volume
- sell_aggressor_volume
- cancel_to_add_ratio
- near_touch_cancel_pressure
- top_1_depth
- top_3_depth
- top_5_depth
- top_10_depth
- queue_depletion_rate
- refill_speed
- refill_ratio
- spread_stress
- book_slope
- book_slope_change
- absorption_score
- iceberg_reload_score
- liquidity_vacuum_score
- lead_lag_ms
- micro_vs_mini_lag
- rates_equity_impulse_score
- adverse_selection_after_fill
- fill_probability
- cancel_burst_score
- stop_run_score
- forced_flatten_score
- cutoff_pressure_score
- prop_reentry_score
- news_restriction_flatten_score
- realized_latency_state
- volatility_state
- liquidity_state
- regime_state

17. Opportunity Registry - 44 Candidate L3/Microstructure Families

These are hypothesis and feature families. The final model may use them directly, combine them nonlinearly, discard them, or discover alternatives. Each family requires a formal hypothesis, MBO signature, feature set, target labels, action space, latency sensitivity, queue/fill assumptions, failure modes, and research card.

#	Family	MBO signature
1	Second-wave continuation	Initial event sweep followed by continued same-side aggressor flow.
2	Stop-run exhaustion fade	Sweep through obvious level, stop-flow spike, aggressor deceleration, opposite-side refill.
3	Liquidity vacuum continuation	Top 3-10 book levels vanish and fail to refill.
4	Depth-refill imbalance	After shock, one side rebuilds faster and more persistently.
5	Spread blowout/recompression	Spread widens violently, then compresses with improving depth.
6	Aggressor deceleration fade	Market orders continue but each burst creates less price movement.
7	Forced liquidation cascade	One-way aggressive flow, widening spread, repeated queue depletion, weak refill.
8	False breakout trap	Price breaks prior high/low, but MBO follow-through fails.
9	Cancel storm before move	Large near-touch cancels before price breaks.
10	Queue depletion trigger	Best bid/ask queue drains faster than refill.
11	Book slope collapse	Depth disappears across top 3-10 levels.
12	Absorption fade	Aggressors repeatedly hit same level; price fails to move.
13	Iceberg/reload detection	Displayed level repeatedly trades and replenishes.
14	Liquidity defense break	Large defended level stops reloading.
15	One-sided add/cancel imbalance	Adds stack one side while cancels remove the opposite side.
16	ES to MES lead-lag	ES order-flow pressure leads MES reaction.
17	NQ to MNQ lead-lag	NQ order-flow pressure leads MNQ reaction.
18	ES/NQ divergence snapback	Short-term ES/NQ correlation gap opens and reverts.
19	ZN/ZB to ES/NQ macro impulse	Rates futures react first around CPI/FOMC; equity futures respond after.
20	Micro contract retail lag	Main contract moves first; micro contract reacts slower or noisier.
21	Round-number stop sweep	Stop flow near obvious handles, prior highs/lows, premarket highs/lows.
22	Prior high/low breakout trap	Breakout with weak depth/refill/aggressor confirmation.
#	Family	MBO signature
23	Opening candle chase	09:30 ET impulse followed by late retail chase or reversal.
24	VWAP defense/break	Repeated reload near VWAP, followed by defense hold or failure.
25	DOM illusion trap	Large displayed size cancels before price reaches it.
26	Late candle entry fade	Visible candle expansion with weakening underlying aggressor flow.
27	Stop-loss cascade continuation	Obvious stops convert into market orders and book remains thin.
28	Panic market-order spread tax	Retail/forced orders cross widened spread during stress.
29	End-of-day forced flatten flow	Directional exit pressure into known forced-flat windows.
30	Cutoff panic exits	Urgent flattening in final minutes before prop-firm cutoff.
31	No-overnight inventory squeeze	Late-day one-way reduction of open directional exposure.
32	Daily loss-limit defense	Sudden flattening after adverse intraday move.
33	Trailing drawdown pressure	Panic exit after reversal threatens account state.
34	Profit-lock behavior	Trend-day winners close into cutoff/payout-protection window.
35	Max-contract crowding in MES/MNQ	Micros show noisier forced flow than ES/NQ.

#	Family	MBO signature
36	Prop reset/reopen window	Re-entry behavior after allowed session resumes.
37	Friday/weekend de-risking	Friday final-hour exposure reduction.
38	Economic-event restriction flattening	Pre-news flattening and post-news re-entry around restricted news windows.
39	Quote pull before volatility	Depth vanishes before scheduled high-impact events.
40	Re-quote race after shock	Depth returns unevenly after event shock.
41	Thin-book continuation	Market makers remain wide after first move.
42	Passive trap fill	Passive fills followed by immediate adverse selection.
43	Rebate trap avoidance	Maker-style fills lose more to adverse selection than spread/rebate benefit.
44	Spread regime change	Spread remains elevated longer than baseline after shock.

18. Model Validation Standard

Required validation:

- walk-forward testing only
- event-family separation
- product separation
- latency-band testing
- fee/slippage testing
- queue/fill realism
- adverse-selection testing
- multiple-testing control
- deflated Sharpe or equivalent multiple-test penalty
- regime robustness
- model stability
- confidence calibration
- leakage audit
- timestamp audit
- p95/p99 latency sensitivity
- degradation analysis
- live/sim disagreement report

Latency bands:

0.5ms, 1ms, 2ms, 5ms, 10ms

Metrics:

net PnL, expectancy, Sharpe, deflated Sharpe, Sortino, max drawdown, hit rate, average win/loss, profit factor, fill rate, missed-fill rate, cancel rate, reject rate, adverse selection at 100ms/500ms/1s/5s, tail loss, expected shortfall, p95/p99 latency sensitivity, turnover, capacity estimate, confidence calibration.

19. Position Sizing Contract

Position size must be generated mathematically and configuration governed.

```
size_t = min(
    liquidity_limit,
    margin_limit,
    expected_shortfall_limit,
    daily_loss_limit,
    confidence_scaled_size,
    latency_adjusted_size,
    product_max_size,
    strategy_or_model_max_size
)
```

Required outputs:

raw model size, liquidity-capped size, risk-capped size, margin-capped size, final approved size, reason for any reduction, expected loss if stopped, expected shortfall contribution.

20. Production State Machine

Model deployment is evidence-driven. Transitions and thresholds must be configuration-defined, unit-labeled, versioned, and visible in reports.

State	Meaning	Allowed next states
RESEARCH	Hypothesis/model under construction.	BACKTESTED, REJECTED
BACKTESTED	Historical replay completed with reports.	SIM_SHADOW, RESEARCH, REJECTED
SIM_SHADOW	Runs against Rithmic sim and produces execution-truth comparison.	MICRO_LIVE, BACKTESTED, REJECTED
MICRO_LIVE	Smallest live production exposure for evidence gathering.	SCALE_REVIEW, QUARANTINED
SCALE_REVIEW	Evidence packet reviewed for scale approval.	ACTIVE, MICRO_LIVE, QUARANTINED
ACTIVE	Production-active under current configuration.	QUARANTINED, RETIRED
QUARANTINED	Blocked from new entries pending review.	RESEARCH, RETIRED, ACTIVE
RETIRED	Disabled but retained for audit/history.	RESEARCH

21. Production Database Schemas

Implement durable tables or append-only logs with these minimum fields. Parquet is acceptable for research logs; a relational DB or event store is acceptable for production state.

model_decisions

decision_id, session_id, timestamp, product, model_id, model_version, state_id, action, expected_gross_edge, expected_cost, expected_net_edge, fill_probability, adverse_selection_estimate, latency_band, confidence_score, risk_score, requested_size, approved_size, risk_status, config_hash

execution_events

event_id, decision_id, order_id, product, side, order_type, price, quantity, order_send_ts, ack_ts, fill_ts, cancel_send_ts, cancel_ack_ts, replace_send_ts, replace_ack_ts, reject_ts, status, reject_reason, rithmic_session_id

execution_truth

truth_id, decision_id, expected_fill_probability, actual_fill, expected_slippage, actual_slippage, expected_adverse_selection, actual_adverse_selection_100ms, actual_adverse_selection_500ms, actual_adverse_selection_1s, expected_latency, actual_latency, expected_net_edge, actual_net_edge

risk_events

risk_event_id, timestamp, session_id, product, model_id, risk_check, result, reason, open_orders, position, realized_pnl, unrealized_pnl, action_taken

session_health

timestamp, rithmic_status, data_status, clock_offset, p50_latency, p95_latency, p99_latency, disconnect_count, reject_count, stale_data_flag, position_match_flag

research_cards

card_id, model_id, hypothesis_family, years_tested, events_tested, symbols_tested, features_used, targets_used, queue_model, latency_model, gross_pnl, net_pnl, drawdown, fill_rate, adverse_selection, expected_shortfall, pass_fail, recommendation

databento_requests

request_hash, event_id, event_type, symbols, start_utc, end_utc, duration_seconds, cost, cumulative_cost, remaining_credit, resolved_contract, instrument_id, file_path, status

22. Daily Operating Workflow

Phase	Required outputs
Pre-session	Server health, Rithmic login, clock sync, config hash, product enablement, risk config, stale data check, emergency contact availability.
During session	Latency monitor, order/fill monitor, data freshness monitor, position reconciliation, rejection monitor, risk events, operator alerts.
Event window	Event calendar lock, model state snapshot, decision logs, execution truth capture, anomaly flags.
Post-session	Daily PnL reconciliation, fill-quality report, adverse-selection report, latency report, reject report, sim-live divergence report, research feedback packet.

23. Production Failure States

Implement hard failure states with durable logs and operator alerts:

```
STALE_MARKET_DATA
RITHMIC_DISCONNECT
CLOCK_DRIFT
LATENCY_SPIKE
MISSING_FILL_RECONCILIATION
ORDER_REJECT_CLUSTER
POSITION_MISMATCH
MARGIN_OR_RISK_WARNING
DAILY_LOSS_HALT
MANUAL_KILL_SWITCH
EMERGENCY_FLATTEN
```

Each failure event must record:

```
reason, timestamp, open orders, current position, attempted action,
action taken, operator alert, recovery requirement, config hash.
```

24. Deliverables for Quote

Package	Deliverables
Package 1 - Bare-Metal Server Build	Linux install, CPU isolation, process pinning, IRQ tuning, clock sync, security hardening, Rithmic connectivity test, latency report.
Package 2 - Rithmic Gateway	Market data listener, order entry, cancel/replace, fills/rejects, reconnect, heartbeat, position state, emergency flatten, latency logger.
Package 3 - Databento Research Data System	Event calendar builder, cost estimator, budget-controlled downloader, manifests, cost-per-symbol-minute report, duplicate prevention, UTC validation, roll metadata.
Package 4 - HftBacktest Pipeline	Databento conversion, queue models, latency bands, empirical latency support, replay runner, model plug-in interface, fee/slippage model.
Package 5 - Feature and Opportunity Research	Core MBO features, all 44 hypothesis families as testable modules, reports, research cards, ranking board, latency sensitivity board.
Package 6 - Model Training and Decision Engine	Model-agnostic training, black-box support, target construction, leakage controls, walk-forward validation, artifact registry, decision object output, calibration.
Package 7 - Production Operations	State machine, execution-truth schema, reconciliation, session health, dashboards, operator alerts, daily reports.
Package 8 - Risk and Readiness	Risk checks, halts, kill switch, stale-data halt, disconnect halt, position mismatch handling, sim-to-live checklist.

25. Acceptance Tests

- Bare-metal server produces reproducible latency, jitter, packet-loss, and clock-sync reports.
- Rithmic sim/live connectivity works from the server and produces heartbeat/session logs.
- Databento downloader reproduces every event window from events.csv and enforces cost controls.
- All raw requests include cost, UTC conversion, resolved contract, instrument ID, request hash, and file path.
- HftBacktest conversion works from raw .dbn.zst to .npz and produces conversion manifests.

- All features are timestamp-safe and leakage-audited.
- All 44 opportunity families are represented as research modules and research cards.
- Model training supports transparent and black-box models with reproducible artifacts.
- Latency bands 0.5ms, 1ms, 2ms, 5ms, and 10ms are tested.
- Empirical Rhythmic latency model is produced after sufficient samples exist.
- Models are ranked by net expectancy, fill quality, latency sensitivity, adverse selection, and tail risk.
- State transitions are logged with config hash and evidence packet.
- Risk layer can block, flatten, and halt; each failure state produces an operator alert and durable log.
- Daily post-session report reconciles broker/FCM truth, Rhythmic fills, local position state, and model expectations.
- Final report shows Databento credit used, remaining credit, and forecast expansion cost.

26. Quote Request

Provide fixed-price and hourly estimates by package, delivery timeline, required credentials, server/Rhythmic/Databento access needs, external dependencies, known risks, test plan, production-readiness checklist, and maintenance estimate after delivery. The quote must assume a professional-grade Chicago futures microstructure platform with research, model decisioning, production operations, and execution-truth validation as one integrated system.

27. Source Anchors

Reference materials to verify during implementation:

- Databento documentation: CME Globex MDP 3.0 / GLBX.MDP3, MBO schema, Historical API metadata.get_cost, continuous futures symbology.
- HftBacktest documentation: Databento conversion utilities, queue models, order-fill and latency models.
- BLS release calendars: CPI and Employment Situation release times.
- Federal Reserve FOMC calendars and statement/press-conference schedules.
- Prop-firm public rules pages for Topstep, Apex Trader Funding, MyFundedFutures, and Take Profit Trader windows.
- Mathematical foundation: probability, stochastic processes, statistics, econometrics/time series, numerical methods, optimization/control, finance, and computer systems as core disciplines for production-grade quantitative finance research.